

JCars Logistics Power BI Business Intelligence Assessment

From Raw Data to Business Decisions

1. Assessment Brief

JCars Logistics imports, sells, and delivers vehicles to customers across different regions in Kenya. The company collects information about its sales transactions, vehicles, customers, branches, sales representatives, payments, deliveries, logistics costs, returns, cancellations, and customer experiences. Management would like to use this information to better understand how the business is performing and identify areas that require attention. You have been provided with a raw flat dataset containing JCars Logistics sales and operational records. The dataset has not been prepared for analysis and may contain data quality issues, inconsistencies, missing information, unusual values, formatting problems, and other issues that could affect the reliability of management reporting. You have been engaged as a Business Intelligence and Data Analyst to transform this raw dataset into a reliable and interactive Power BI business intelligence solution. Your responsibility is not simply to create charts. You are expected to understand the business and the dataset, investigate the quality of the data, identify and resolve appropriate data problems, document assumptions, prepare the data using Power Query, design an appropriate analytical data model, develop meaningful DAX calculations, validate your work, create a one-page Executive Dashboard, develop detailed analytical report pages, investigate unusual business patterns, generate insights, and provide evidence-based recommendations to management. Your final solution should demonstrate your ability to move from raw data to useful business information and ultimately to management decision support.

2. Business Objective

The objective of this assessment is to develop a Power BI solution that enables JCars Logistics management to understand the overall performance of the business and investigate the factors contributing to that performance. Management should be able to use your solution to understand sales and revenue performance, costs and profitability, vehicle performance, customer performance, branch and geographic performance, sales representative performance, sales channels and lead sources, payment performance, delivery and logistics performance, returns and cancellations, customer experience, changes in performance over time, and unusual transactions or business areas that may require further investigation. Your analysis should go beyond reporting numbers. Where possible, your work should help management understand what happened, where it happened, how significant it was, and what deserves further attention.

3. Understanding the Dataset

Before cleaning, transforming, modelling, or visualising the data, study the dataset carefully and determine what it represents. Identify what one row represents and clearly state the grain of the dataset. Examine the available columns and determine what business information they contain. Consider which fields describe customers, vehicles, locations, branches, sales representatives, transactions, payments, logistics activities, dates, and other business entities. Determine which fields represent measurable business values and which fields are descriptive attributes. Identify

fields that could potentially be used as identifiers or keys and consider whether they are actually unique. You should also begin thinking about which fields may eventually belong in fact and dimension tables. Do not begin building the dashboard before you understand the data. Your project documentation should include a short explanation of the dataset and its grain.

4. Data Quality Audit

The dataset has intentionally been provided in raw form. You should not assume that every value is complete, consistent, correctly formatted, correctly classified, or suitable for immediate analysis. Conduct a comprehensive data quality audit before developing your analytical solution. Investigate the dataset for missing information, inconsistent formatting, inconsistent categories, spelling variations, incorrect or inconsistent data types, duplicate or potentially duplicate records, suspicious numerical values, inconsistent dates, possible outliers, inconsistent monetary values, values that may violate reasonable business rules, and any other problems that could affect the reliability of the analysis. This list is not exhaustive. Part of the assessment is your ability to discover problems independently. Document at least 10 significant data quality issues that you identify. For each issue, explain what you found, why it represents a potential problem, how it could affect the analysis, how you decided to handle it, and any assumptions you made. You are responsible for deciding whether an issue should be corrected, standardized, replaced, excluded, flagged, retained, or investigated further. Do not remove a record simply because it appears unusual. Where an issue cannot be corrected confidently using the available evidence, make a reasonable analytical decision and document it.

5. Currency Standardization Challenge

JCars Logistics operates primarily in Kenya, but the dataset contains monetary values that may have been recorded using different currencies and currency formats. Your final management analysis must present comparable monetary values using Kenya Shillings, KES or KSh, as the reporting currency. Investigate the dataset and identify the currencies that appear in the monetary fields. Determine which monetary columns are affected and decide how the different currencies should be identified and standardized before financial calculations are performed. Where a monetary value does not explicitly indicate a currency, assume that the value is in Kenya Shillings. Where another currency is explicitly indicated, the value must be appropriately converted before it is combined or compared with Kenya Shilling values. You are responsible for determining appropriate exchange rates, identifying a credible source for those rates, documenting the rates used, and explaining your currency conversion approach. You should use one consistent set of exchange rates for the project unless you provide a reasonable justification for another approach. The assessment is not testing whether everyone obtains exactly the same converted value. It is testing whether you identify the currency problem, handle it consistently, document your assumptions, and ensure that financial analysis compares values expressed on a common basis.

6. Data Cleaning and Preparation Using Power Query

Use Power Query to transform the raw dataset into reliable analysis-ready data. Your transformations should be informed by the problems discovered during your data quality audit

rather than performed mechanically. Consider the appropriate treatment of missing values, inconsistent categories, text formatting, dates, numerical fields, currencies, data types, duplicates, errors, and other identified problems. You may create additional fields where they are analytically justified. You should be able to explain why important transformations were performed. Do not change a value merely because it looks unusual. Where the available information does not provide sufficient evidence to determine the correct value, decide whether the record should be retained, flagged, excluded from a particular calculation, or treated in another appropriate manner. Your approach should preserve the integrity of the original business information while producing data that can support reliable analysis.

7. Data Validation

Cleaning data does not automatically mean that the resulting data is correct. After completing your transformations, validate the analysis-ready data. Confirm that important fields have appropriate data types, categories have been standardized where necessary, monetary values used in common calculations are represented consistently, important records have not accidentally disappeared, transformations have not unintentionally duplicated business values, and the resulting dataset remains consistent with the business scenario. Investigate whether values already recorded in the dataset can be independently checked using other available fields. Where calculated values and recorded values do not agree, investigate the difference rather than automatically choosing one of them. Document important validation checks and explain any significant discrepancies discovered.

8. Data Modelling

The supplied dataset is a raw flat table. Design an appropriate analytical data model that can support your Power BI analysis. Determine whether the original flat structure is appropriate for your final analytical solution or whether the data should be organised into multiple related tables. Identify the main business process being analysed and determine which table or tables should represent facts and which should represent dimensions. Determine appropriate keys and relationships between your tables. Configure appropriate relationship cardinality and cross-filter direction. Consider how dates should be handled to support reliable time-based analysis. Your model should avoid unnecessary duplication and ambiguous relationships while supporting the business questions in this assessment. You are expected to apply the principles of data modelling, relationships, fact tables, dimension tables, cardinality, filtering, and analytical schemas covered in class. You must be able to justify your modelling decisions. Your final Power BI Model View should be organised and understandable, and a screenshot of the completed model must be included in your project documentation.

9. Business Logic and DAX

Develop appropriate DAX measures to support reusable business calculations throughout your Power BI solution. Do not rely only on automatically generated or implicit aggregations. Your calculations should enable management to analyse areas including revenue, costs, profitability, units sold, orders, customers, transaction performance, delivery performance, logistics, returns, cancellations, customer ratings, rankings, contributions to overall performance, and changes in

performance over time. Your work should demonstrate appropriate application of DAX concepts covered in class. Where relevant to your analysis, demonstrate your ability to work with filter context, conditional calculations, ratios, rankings, and time-based calculations. You are responsible for determining which measures are required to answer the business questions. Do not create measures simply to increase the number of DAX calculations in your project. Every measure should have an analytical purpose.

10. Business Definitions and Assumptions

Some business terms in the assessment may have more than one reasonable interpretation. These may include revenue, cost, gross profit, gross profit margin, customer value, sales performance, logistics efficiency, vehicle performance, branch performance, returns, cancellations, and other analytical concepts. Where the dataset or assessment does not explicitly provide a definition, choose a reasonable definition based on the available information, document your assumption, and apply the definition consistently throughout your analysis. Do not make important assumptions silently. The quality of your reasoning and consistency will form part of the assessment.

11. Business Questions

Your Power BI solution should enable JCars Logistics management to investigate the following business questions. What is the total number of cars sold? What is the total sales revenue generated by the company? What is the total gross profit and gross profit margin? Which car makes generate the strongest business performance? Which car models perform strongly based on units sold? How do different vehicle types perform when evaluated using revenue, units sold, and profitability? Which branches generate the strongest business performance? How does performance vary by region, county, or other available geographic levels? How are individual sales representatives performing? Which lead sources or sales channels generate valuable business? How have revenue, units sold, and gross profit changed over time? How do different payment methods contribute to business performance? What is the payment status profile of the business when analysed by both value and transaction volume? What can management learn from the different delivery statuses? How does delivery performance vary across regions, branches, vehicles, or other relevant business categories? Which parts of the business have high logistics costs relative to the value they generate? Which car makes or models generate strong profit margins? Which customer types contribute most to the business? What relationship, if any, exists between discounts and sales performance? What relationship, if any, exists between customer ratings and revenue, units sold, or other relevant measures? Which orders or customers should management investigate further? Who are the company's highest-value customers? Which vehicles appear frequently among returns or cancellations? What can management learn about vehicle performance by year of manufacture? What are the most important findings management should understand from the available data? What business recommendations can you provide based on the evidence from your analysis? These are business questions and not instructions to create individual charts. You should not create one visual for every question. A well-designed visual or report page may answer several related questions.

12. Analyst-Defined Business Questions

A good analyst should be able to identify important questions that were not explicitly requested by management. After exploring the dataset, develop at least five additional business questions that you believe JCars Logistics management should be asking. Your questions must add analytical value and should not simply reword the questions already provided. For each question, explain why it matters to the business and determine how it can be answered using the available data. At least three of your analyst-defined questions must be answered within your Power BI solution. Your ability to identify useful questions from the available data will form part of the assessment.

13. Investigation and Exceptions

Your analysis should help management identify transactions, customers, vehicles, branches, sales representatives, logistics activities, or other business areas that may require further investigation. You are responsible for determining what constitutes an unusual or potentially concerning observation. Your investigation may consider patterns involving sales, costs, discounts, profitability, logistics, delivery performance, payment status, returns, cancellations, customer ratings, transaction values, or other relevant variables. Do not label something as a problem simply because it is different from other observations. Use evidence from the data. Where you establish a threshold, classification, or business rule for identifying unusual observations, explain your reasoning. Your objective is to demonstrate analytical judgement rather than simply locate the largest or smallest number in a column.

14. Executive Dashboard

The first page of your Power BI solution must be one Executive Dashboard. This should be a single-page management overview that enables JCars Logistics management to quickly understand the overall state of the business. The Executive Dashboard should broadly answer the question, "How is JCars Logistics performing?" The dashboard should summarize the most important information from your analysis and should contain an appropriate combination of key performance indicators, business trends, performance comparisons, important business breakdowns, useful slicers or filters, and indicators of areas that may require management attention. You are responsible for deciding which KPIs deserve executive attention, which trends should be immediately visible, which comparisons are important, which filters are useful, and which visuals communicate the information most effectively. Do not attempt to display every available measure or answer every business question on this page. The purpose of the Executive Dashboard is summary and monitoring rather than detailed investigation. Selecting what not to include is part of good dashboard design. The dashboard should fit logically on one Power BI report page, have a clear title and visual hierarchy, display important information prominently, use appropriate number and currency formatting, maintain consistent formatting, use appropriate spacing and alignment, and be understandable without requiring the analyst to explain every visual. A manager viewing only this page should be able to develop an initial understanding of the company's performance.

15. Detailed Analytical Report

After the Executive Dashboard, create detailed report pages that allow management to investigate the business in greater depth. The Executive Dashboard should provide the high-level view, while the detailed report pages should help management understand where, how, and potentially why particular patterns are occurring. Your detailed report should support deeper investigation of sales and revenue, costs and profitability, vehicles, customers, branches and geographic performance, sales representatives and sales channels, payments, logistics and deliveries, returns and cancellations, customer experience, trends over time, rankings and comparisons, and unusual transactions or business areas requiring investigation. You are not required to create one report page for every analytical area. Related analyses may be combined where this produces a clearer and more useful report. You are responsible for determining the appropriate number of pages, the purpose and name of each page, the KPIs required on each page, the most appropriate visuals, visual placement, filters, slicers, interactions, navigation, and information hierarchy. The business questions in this assessment guide what should be investigated but do not prescribe how your report should look.

16. Dashboard and Report Analytical Flow

Your Executive Dashboard and detailed report pages should function as one analytical solution rather than separate collections of visuals. The intended analytical flow should allow management to begin with the Executive Dashboard, identify an important trend, issue, or opportunity, move into the detailed report, investigate the underlying information, develop a deeper understanding of the pattern, and reach an evidence-based business conclusion. You are responsible for deciding how this analytical experience should be implemented. A strong solution should make it easy for a manager to move from high-level monitoring to detailed investigation.

17. Interactivity

Your Power BI solution should demonstrate that Power BI is an interactive analytical tool rather than simply a presentation tool. Implement meaningful interactivity that assists management in exploring the data. Your solution must include useful slicers or filters, appropriate interactions between visuals, at least one drill-through experience, at least one useful report tooltip, and appropriate navigation between report pages. These features should serve an analytical or business purpose. Do not add features simply to demonstrate that you know how to use them. A drill-through page, tooltip, slicer, button, or navigation feature should make it easier for the user to answer a question or investigate an observation.

18. Visual Selection

You are responsible for selecting the most appropriate visuals for your analysis. Depending on the question being answered, you may determine that a KPI, trend visual, comparison visual, ranking, geographic visual, table, matrix, relationship visual, detailed investigation visual, or another appropriate Power BI visual is suitable. You are not being told which chart should answer each business question. Choosing an appropriate visual is part of the assessment. Avoid using a visual simply because it has not been used elsewhere in your report. Every visual should

have a clear analytical purpose. One appropriately designed visual may answer several related business questions.

19. Dashboard and Report Design

Your complete Power BI solution should demonstrate professional dashboard and report design principles. Consider visual hierarchy, titles, labels, number formatting, currency formatting, alignment, spacing, readability, consistency between pages, use of report space, navigation, and overall presentation. Avoid overcrowding report pages. Important information should be easy to identify. Your use of colours, formatting, and visual emphasis should support interpretation rather than distract from the information. Monetary values displayed together should use a consistent reporting currency after currency standardization. Every visual should contribute to management's ability to understand, compare, monitor, investigate, or make decisions about the business.

20. Management Insights

After completing your analysis, provide at least five meaningful business insights. An insight should go beyond simply repeating a value displayed on a visual. For example, stating that a particular branch generated a particular amount of revenue is primarily an observation. A stronger insight provides context by explaining what happened, where or when it happened, how significant the pattern is, what other information helps explain it, and whether it deserves management attention. Your insights must be supported by evidence from your analysis. Where the available data does not provide enough evidence to explain why something occurred, state that it requires further investigation rather than presenting an unsupported explanation as fact.

21. Management Recommendations

Provide at least three evidence-based recommendations to JCars Logistics management. Each recommendation must clearly connect to a finding from your analysis. Avoid generic recommendations such as "increase sales," "reduce costs," or "improve customer satisfaction" unless your analysis demonstrates specifically where and why management should take action. Your recommendations should identify what management should consider investigating, changing, improving, reducing, continuing, or monitoring based on the evidence available. Be careful not to claim that the data proves something that it cannot establish. A relationship between two variables does not automatically mean that one caused the other.

22. Assumptions and Business Rules

Create an Assumptions and Business Rules section in your project documentation. Document the important decisions you made throughout the project. This should include assumptions relating to currency, missing values, revenue, costs, profit, returns and cancellations, dates, customer value, data corrections, unusual records, and any other ambiguous situations encountered during the project. Where monetary values do not explicitly state a currency, the required assumption for this assessment is that they are in Kenya Shillings. Other assumptions should be determined

and justified by you based on the available evidence. Your assumptions should be applied consistently throughout the project.

23. Final Management Test

Before submitting your project, imagine that a JCars Logistics manager opens your Power BI solution without you being present. On Page 1, the Executive Dashboard should allow the manager to develop a high-level understanding of how the business is performing, what the most important numbers are, what important trends are developing, which areas are performing differently from others, and whether anything appears to require management attention. The detailed report pages should then allow the manager to investigate where these patterns are occurring, what information may explain them, which underlying records or categories are contributing to them, and what should be investigated further. Your final solution should demonstrate a logical progression from dashboard to detailed report, investigation, insight, and business decision-making.

24. Project Documentation

Your project must be properly documented. Your documentation should explain the project background, business objective, dataset and grain, data quality issues discovered, data cleaning and transformation approach, currency standardization approach, exchange rate assumptions and sources, validation checks, assumptions and business rules, data modelling approach, fact and dimension tables, relationships, important DAX measures, Executive Dashboard design, detailed report structure, interactivity implemented, analyst-defined questions, key findings, management insights, recommendations, challenges encountered, and what you learned from the project. Include appropriate screenshots where they help demonstrate your work. Screenshots should be clear, readable, logically arranged, and accompanied by explanations where necessary.

25. GitHub Submission

Upload your completed project to GitHub. Your repository should contain your Power BI .pbix file, the dataset used for the final analysis, a professional README.md file, relevant screenshots, and any additional supporting documentation required to understand your work. Your README should clearly explain the project objective, dataset, data quality investigation, cleaning and transformation approach, currency handling, assumptions, data model, important DAX calculations, dashboard and report structure, key insights, and management recommendations. Organise the repository professionally. Do not upload passwords, access credentials, API keys, personal authentication information, or other sensitive information.

26. Technical Article

Publish a technical article on Dev.to documenting your JCars Logistics Power BI project. The article should explain your journey from receiving the raw dataset through data investigation, cleaning, preparation, modelling, DAX development, dashboard and report development, analysis, insights, and recommendations. Include clear and well-organized screenshots showing important stages of your work. Your article should explain your reasoning rather than simply

present a sequence of screenshots. A reader should be able to understand what problems existed in the raw data, how you discovered them, how you handled them, what assumptions you made, why you selected your modelling approach, how important calculations support the analysis, how you designed the dashboard and detailed report, and how your final solution supports business decision-making.

27. Submission Checklist

Before submitting, confirm that you have completed the data quality audit, cleaned and prepared the data using Power Query, identified and standardized currencies, documented exchange rates and assumptions, validated your cleaned data, developed an appropriate analytical data model, configured relationships correctly, developed the necessary DAX measures, created one Executive Dashboard as Page 1, created detailed analytical report pages from Page 2 onwards, implemented slicers and appropriate visual interactions, implemented at least one drill-through experience, implemented at least one useful report tooltip, provided appropriate report navigation, answered the required management questions through your analysis, developed at least five additional analyst-defined business questions, answered at least three of your analyst-defined questions in the Power BI solution, investigated unusual records or business patterns, developed at least five meaningful business insights, provided at least three evidence-based management recommendations, documented your assumptions and business rules, prepared a professional GitHub repository and README, and published your technical article on Dev.to.

28. Important Assessment Guidance

There is no single correct dashboard, report structure, data model, set of visuals, or analytical approach for this assessment. Different analysts working with the same dataset may make different reasonable decisions. You will therefore not be assessed only on whether you obtained a particular number. You will also be assessed on the quality of your data investigation, cleaning decisions, Power Query transformations, data modelling, DAX, validation, visual selection, dashboard design, report organisation, analytical reasoning, documentation, insights, and business communication. Do not make a decision simply because you think it is what the instructor expects. Make decisions that you can justify using the data, business context, and Power BI principles covered in class.

Final Challenge

Imagine that you have five minutes with the management team of JCars Logistics. They are not primarily interested in hearing every Power Query transformation you performed or every DAX function you used. They want to understand how the business is performing, where the company is generating value, where it may be experiencing inefficiencies, which customers, vehicles, branches, sales representatives, or sales channels matter to the business, what unusual patterns deserve attention, and what management should investigate or consider doing next. Your Power BI solution should allow you to answer those questions confidently within a short time using evidence from the data.

25. Submission Instructions

The deadline for this assessment is **Saturday, 26 September 2026 at 4:00 PM EAT**.

Each student must submit **two links**:

1. **GitHub Repository Link** containing the completed Power BI project and supporting documentation.
2. **Dev.to Article Link** containing the published technical article documenting the project.

Submit both links by email according to your track:

Data Science Students:

ch9datascience@luxdevhq.com

Data Engineering Students:

ch9dataengineering@luxdevhq.com

GitHub Repository

Your GitHub repository should contain:

- The completed Power BI .pbix file
- The dataset used for the final analysis
- A professional README.md
- Relevant screenshots
- Any additional supporting documentation required to understand your work

Your README should clearly document the project objective, dataset, data quality issues identified, cleaning and transformation approach, currency standardization, assumptions and business rules, data model, important DAX measures, Executive Dashboard, detailed report pages, key insights, and management recommendations.

Do not upload passwords, credentials, API keys, authentication information, or other sensitive information to GitHub.

Dev.to Article

Publish a technical article on **Dev.to** documenting your JCars Logistics Power BI project.

Your article should explain your journey from the raw dataset through data investigation, cleaning, preparation, modelling, DAX development, Executive Dashboard and report development, analysis, insights, and recommendations.

Include clear and well-organized screenshots where appropriate. Your article should explain your reasoning and decisions rather than simply present screenshots of completed work.

What to Submit by Email

Your submission email should contain:

- Your full name
- Your cohort and track
- GitHub repository link
- Dev.to article link

Submission Deadline: Saturday, 26 September 2026 at 4:00 PM EAT

Ensure that both links are accessible before submitting. Inaccessible, private, incorrect, or incomplete links may prevent your work from being reviewed.

Have a wonderful break

